mandate.

Owner-approved business bill payments

An AI wallet agent with independent contract checks

Team ChainSentinel

International Track 2026 · Topic F: Agent Wallet

Synthetic bills and MDT test tokens

The complete payment workflow

First: the owner funds the vault and sets vendors, rules and agent access

Stage	What happens
1 Upload and review	Owner checks source fields and the trusted recipient
2 Sign bill approval	Wallet signature stores exact permission on-chain
3 Plan and confirm	AI request and integer checks produce an exact plan. The user separately confirms execution.
4 Pay and verify	Agent submits. Contract checks. Matching receipt confirms.

Approval grants permission. A confirmed plan starts execution.

The owner can pause or revoke access, effective after chain confirmation

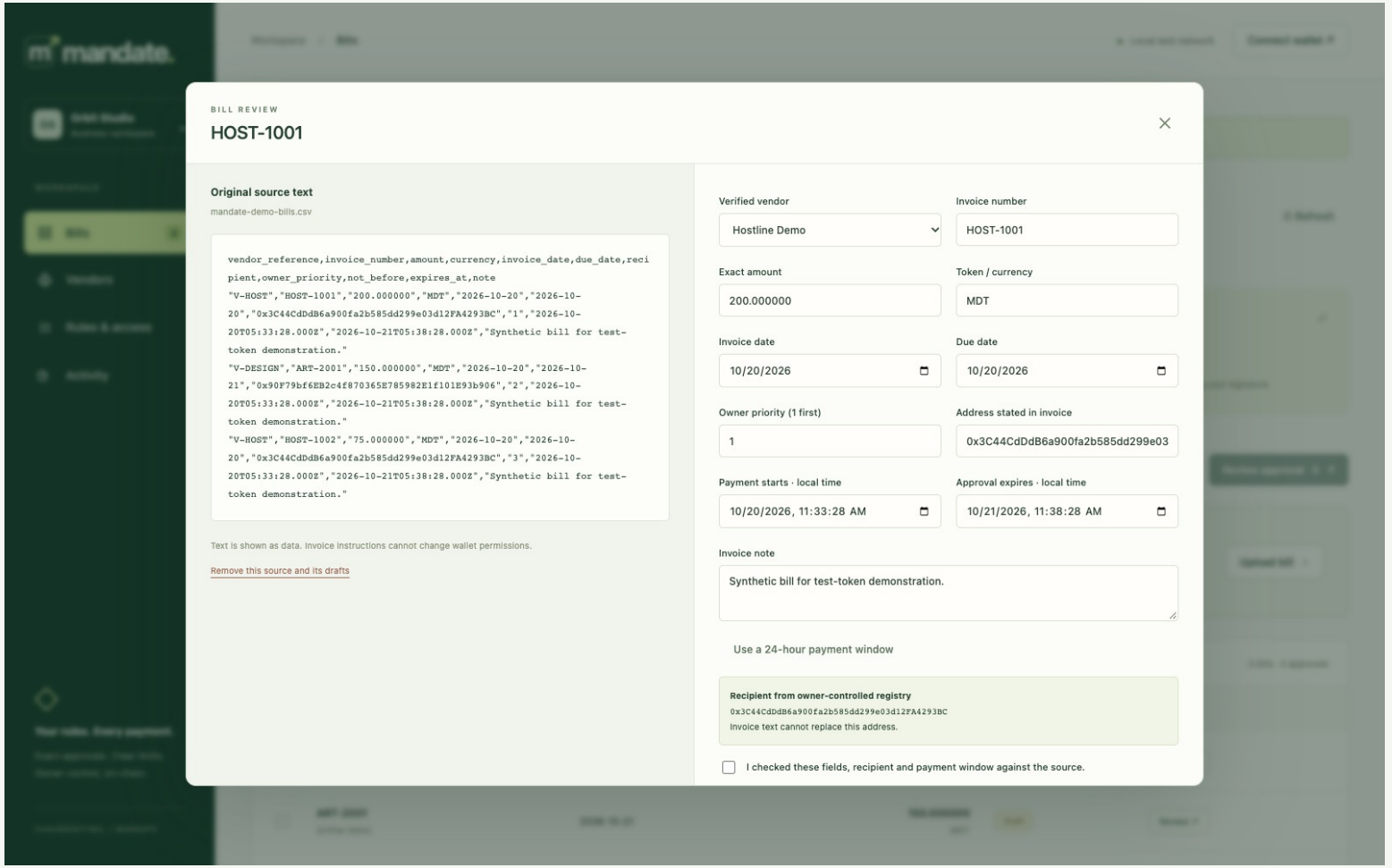
The owner’s bill review

A batch of supplier bills needs exact permission

Compare the source with the draft fields

Use the owner’s trusted vendor recipient

Sign the exact approval before planning



Recorded local scripted replay · chain 31337 · synthetic bills

Payment authority

Component	Job	Payment authority
Owner / wallet	Approve exact bills and rules	Owner signature
AI model	Suggest fields and interpret requests	No keys or approval tools
Application / agent	Plan in integer units, then submit	Confirmed approved bill keys
MandateVault	Check rules and transfer MDT	Stored recipient and amount

Every payment refers to an owner-approved on-chain bill

A focused clarification

OWNER

“Pay that one.”

MANDATE

“Which bill would you like to pay? Please specify the bill ID or invoice number.”

0 selected bills 0 payment attempts

A valid payment plan still needs a separate confirmation

Saved live Groq interaction · 19 October 2026 · Sepolia workspace

The recorded Sepolia payment batch

Bill	Amount MDT	Outcome on 19 October
HOST-1001	200	Paid, matching receipt
ART-2001	150	Paid, matching receipt
HOST-1002	75	Deferred by the daily limit

1,000 starting − 350 paid = 650 MDT remaining

250 MDT per bill · 400 per UTC day · 300 minimum reserve

Owner revocation confirmed at policy version 7

Public testnet · chain 11155111 · test tokens with no cash value

Contract checks and recovery

Before a transfer

Approved bill and vendor

Caps, reserve and active window

Current agent and policy version

After a send

Match the receipt and payment event

Keep paid bill keys consumed

Reconcile the original hash on uncertainty

A timeout never triggers a blind second payment

Public rejection checks used unmined calls. Independent stop isolation is local.

Recorded local evaluation

Measure	Observed result
Ordinary task completion	19 / 20
Exception explanations	20 / 20
Interaction cases, within the 40	6 / 6
Wrong payment cases	0 / 40
Exact critical fields, 26 CSV invoices	156 / 156

Known safe miss: N13 asked an unnecessary extra question

40 fixed synthetic cases · repaired v5 regression · 19 October 2026

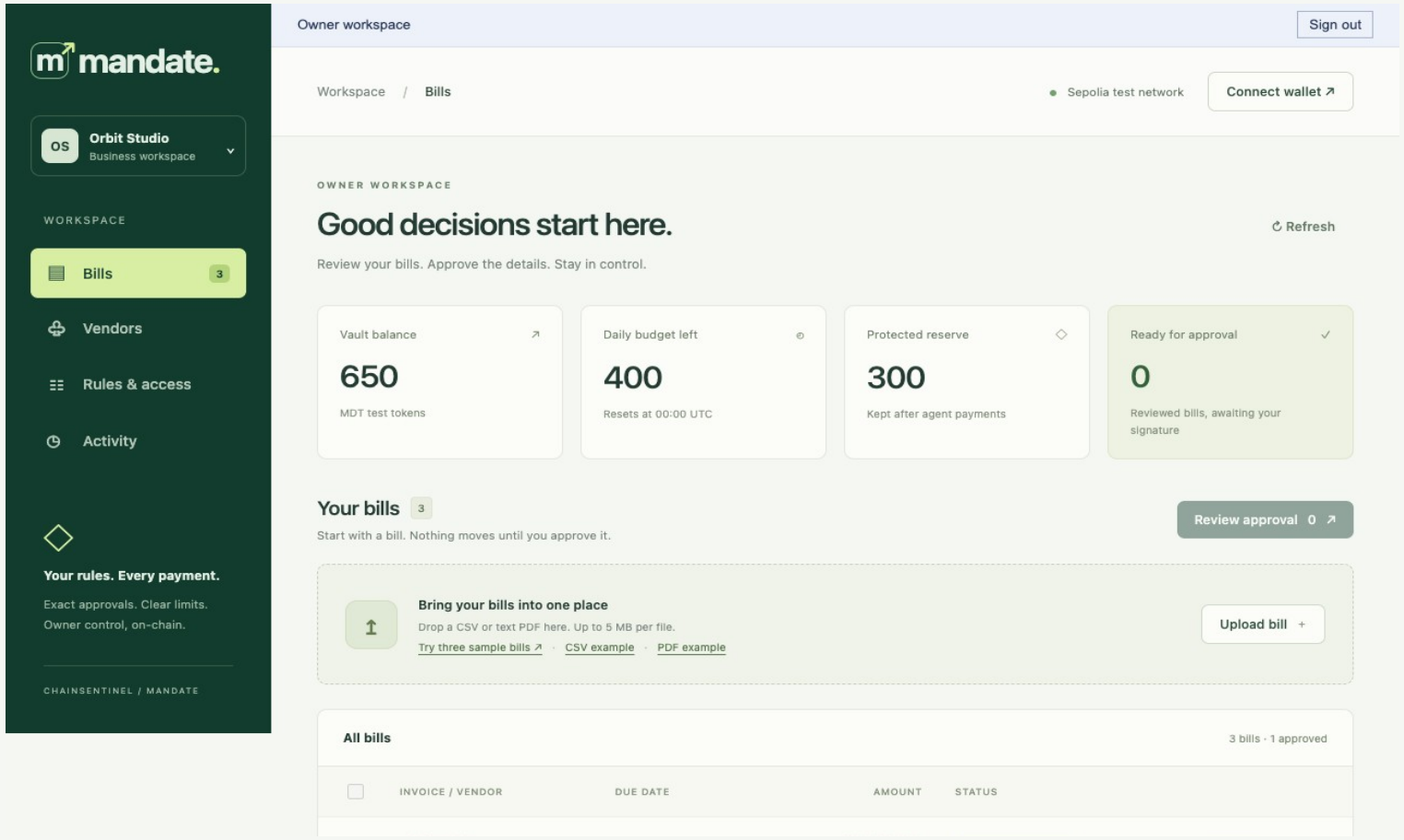
The hosted owner workspace

Render serves the protected application

Neon stores encrypted recovery snapshots

Verified restart preserves bills and receipts

Shared login grants full workspace access



Saved hosted screenshot · 20 October 2026 · existing Sepolia history

Scope and reviewer access

A complete bill workflow with inspectable permission and receipts

Known answer miss: 350 eligible reported instead of 425 approved

Prototype signing storage, shared access and no external audit

No measured human time savings or held-out accuracy claim

Source ZIP and setup README · Execution Evidence and receipt links

https://chainsentinel-mandate.onrender.com

Local development and Sepolia only. Organizer acceptance of Sepolia is unconfirmed.